

Arya Aji

Civil Engineering · Hydraulic Engineering · Sustainable Infrastructure

Munich, Germany · aryaaji337@gmail.com · [linkedin.com/in/arya-aji-8ba163243](https://www.linkedin.com/in/arya-aji-8ba163243) · [arya-ji.github.io](https://github.com/arya-ji)

PROFILE

M.Sc. Civil Engineering student at TU Munich focusing on sustainable infrastructure and hydraulic engineering. Experience in dam design (DIN 19700), physical hydraulic modelling, IFC-based model checking and urban energy analysis, plus site engineering experience. Looking for a working student role, internship or Master's thesis.

EDUCATION

M.Sc. Civil Engineering, Technical University of Munich <i>Sustainable infrastructure, hydraulic engineering, BIM, urban systems, real estate</i>	04/2024 – present
B.Tech Civil Engineering, APJ Abdul Kalam Technological University, Kerala <i>Final grade 2.3 (German scale)</i>	07/2019 – 06/2023

EXPERIENCE

Site Engineer, Fine Form Architects, Pandalam, India • Supervised construction for compliance with drawings, specifications and safety standards • Prepared quantity take-offs; supported cost estimation, resource planning and progress tracking	07/2023 – 02/2024
Hydraulic Model Engineering, TU Munich Obernach Research Station • Physical model tests (Moosburg, Landquart, Baldeysee): hydraulic jumps, energy dissipation, sediment transport • Lab experiments on weirs, spillways, stilling basins, bottom outlets, cavitation and surge tanks	05/2025, 03/2026 – 04/2026

PROJECTS

Dam & Reservoir Design, Thuringia • Designed a concrete dam to DIN 19700 for water supply and flood protection, integrating the existing Vorsperre Deesbach dam • Catchment analysis, storage zoning, KOSTRA flood design, unit hydrographs and GIS site assessment	TU Munich · 04/2025 – 07/2025
BIM Modelling & IFC Model Checking, Office Building • Built a structured IFC model (40 × 16 m) with semantic attributes; rule-based checks in Solibri for clearances, window ratios, accessibility and areas	TU Munich · 04/2025 – 07/2025
Urban Sustainability Analysis: Paul-Gerhardt-Allee, Munich & Alexandria, Egypt • Assessed mobility, green & blue infrastructure, energy and waste; led the energy analysis (demand, efficiency, scenarios)	TU Munich · 10/2024 – 07/2025
Sustainable Clay Blocks (Bachelor project) • Developed clay blocks with red mud, rice husk and water hyacinth; tested strength, absorption and durability to IS codes	07/2022 – 06/2023

SKILLS

Hydraulics	Hydraulic analysis, flow regimes, sediment transport, physical modelling
BIM & Design	Revit, Solibri, IFC, AutoCAD, STAAD Pro (basic), Primavera P6
Sustainability	LCA, urban energy analysis, EU Taxonomy
GIS, Data & Survey	QGIS, MATLAB, Excel, DGPS, Total Station, Surfer

CERTIFICATIONS

EU Taxonomy and Sustainable Finance (2025) · Geotechnical Investigation & Building Construction (2021) · AutoCAD Civil (2021)